

● EASY

● GEOMETRY AND TRIGONOMETRY

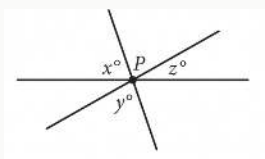
Lines, angles, and triangles

10 questions · ~15 min

01

Q1

GEOMETRY AND TRIGONOMETRY



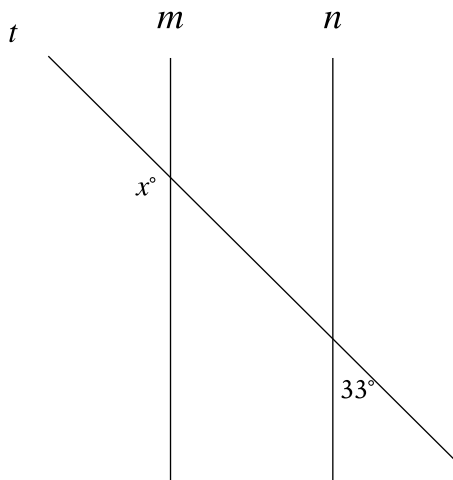
Note: Figure not drawn to scale.

In the figure, three lines intersect at point P. If $x = 65$ and $y = 75$, what is the value of z ?

- A. 140
- B. 80
- C. 40
- D. 20

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 18° . What is the measure of angle F ?

- A. 18°
- B. 72°
- C. 90°
- D. 162°



Note: Figure not drawn to scale.

- Clockwise from top left, the lines are labeled t , m , and n .
- Line t intersects both line m and line n .
- At the intersection of line t and line m , 1 angle is labeled clockwise from top left as follows:
 - Bottom left: x°
- At the intersection of line t and line n , 1 angle is labeled clockwise from top left as follows:
 - Bottom right: 33°
- A note indicates the figure is not drawn to scale.

In the figure, line m is parallel to line n , and line t intersects both lines. What is the value of x ?

- A. 33
- B. 57
- C. 123

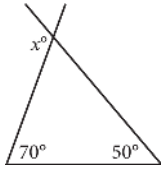
D. 147

At a certain time and day, the Washington Monument in Washington, DC, casts a shadow that is 300 feet long. At the same time, a nearby cherry tree casts a shadow that is 16 feet long. Given that the Washington Monument is approximately 555 feet tall, which of the following is closest to the height, in feet, of the cherry tree?

- A. 10
- B. 20
- C. 30
- D. 35

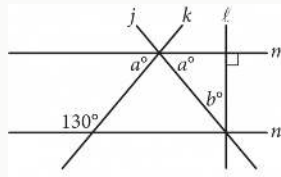
Q5

GEOMETRY AND TRIGONOMETRY



In the figure above, two sides of a triangle are extended. What is the value of x ?

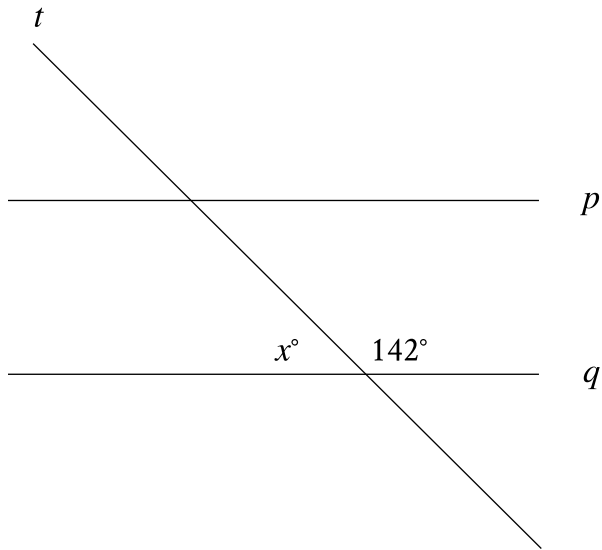
- A. 110
- B. 120
- C. 130
- D. 140



Note: Figure not drawn to scale.

In the figure above, lines m and n are parallel. What is the value of b ?

- A. 40
- B. 50
- C. 65
- D. 80



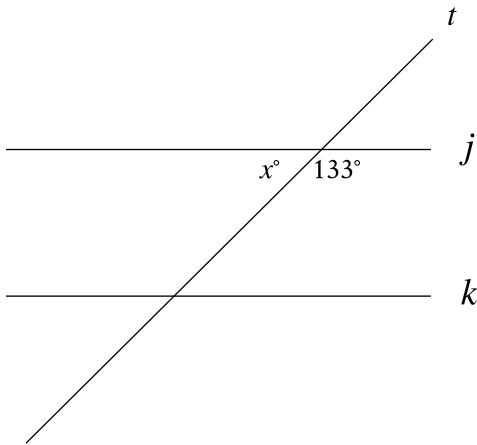
Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t , p , and q .
- Line t intersects both line p and line q .
- At the intersection of line t and line q , 2 angles are labeled clockwise from top left as follows:
 - Top left: x°
 - Top right: 142°
- A note indicates the figure is not drawn to scale.

In the figure, line p is parallel to line q , and line t intersects both lines. What is the value of $x + 142$?

- A. 52
- B. 90
- C. 142

D. 180



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t , j , and k .
- Line t intersects both line j and line k .
- At the intersection of line t and line j , 2 angles are labeled clockwise from top left as follows:
 - Bottom right: 133°
 - Bottom left: x°
- A note indicates the figure is not drawn to scale.

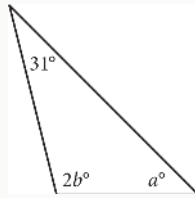
In the figure, line j is parallel to line k . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7

$\frac{\text{ }{\text{8}}{\text{ }}$	$\frac{\text{ }{\text{8}}{\text{ }}$	$\frac{\text{ }{\text{8}}{\text{ }}$	$\frac{\text{ }{\text{8}}{\text{ }}$
$\frac{\text{ }{\text{9}}{\text{ }}$	$\frac{\text{ }{\text{9}}{\text{ }}$	$\frac{\text{ }{\text{9}}{\text{ }}$	$\frac{\text{ }{\text{9}}{\text{ }}$

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



In the triangle above, $a = 45$. What is the value of b ?

- A. 52
- B. 59
- C. 76
- D. 104

In a right triangle, the measure of one of the acute angles is 51° . What is the measure, in degrees, of the other acute angle?

- A. 6
- B. 39
- C. 49
- D. 51